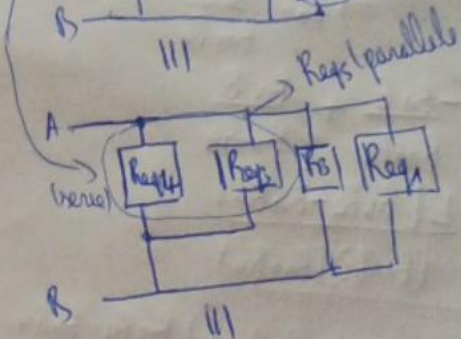
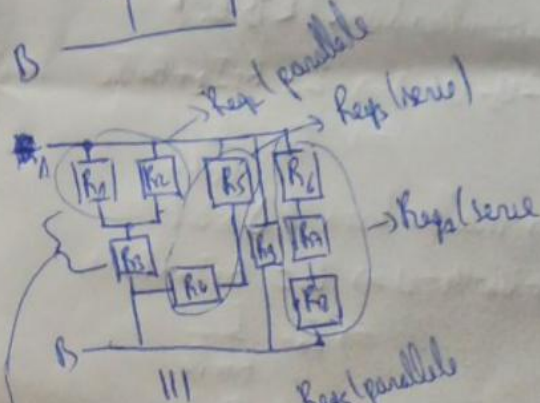
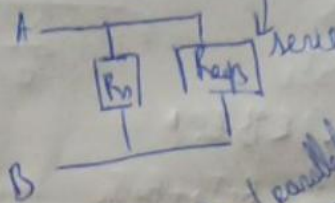
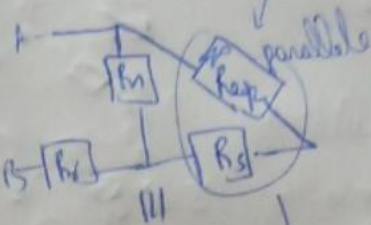
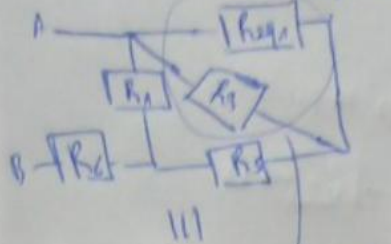
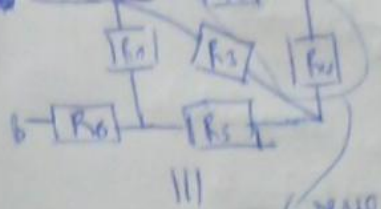
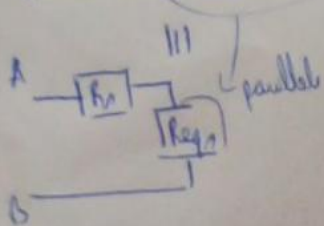
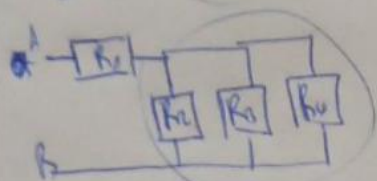
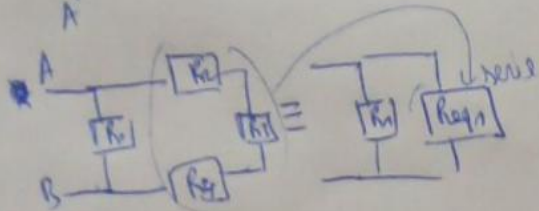
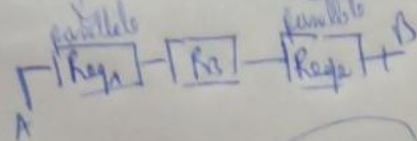
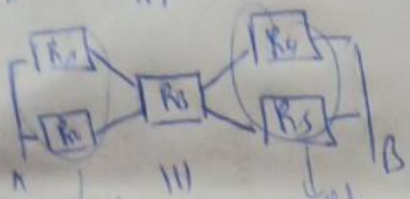
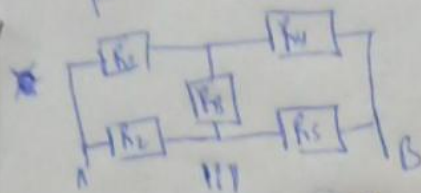
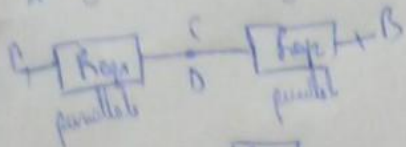
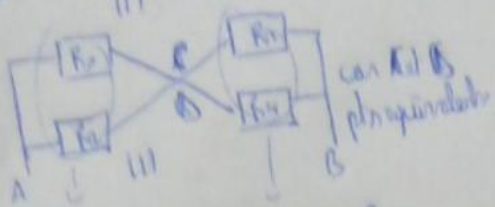
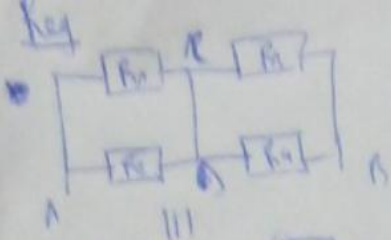
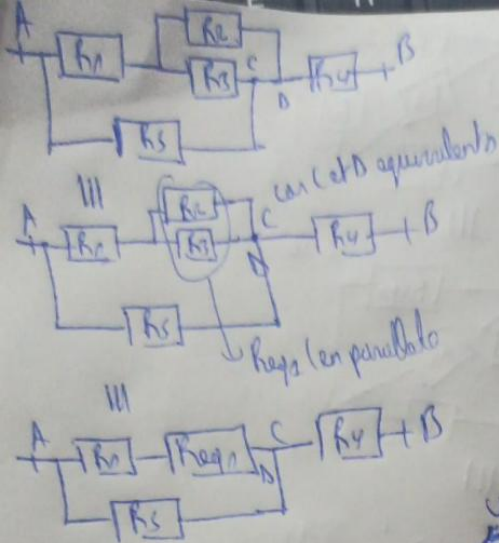






64) (Pont de Wheatstone)

$$U = V_D - V_C = V_D - V_A + V_A - V_C \\ = -U_1 + U_3 \\ = -R_1 I_1 + R_3 I_2$$

- loi des mailles dans A C D B E F A

$$E - U_3 - U_1 = 0 \Rightarrow E = R_3 I_2 + R_1 I_1 \\ \Rightarrow I_1 = \frac{E}{R_1 + R_3}$$

- loi des mailles dans A D B E F A

$$E - U_1 - U_2 = 0 \Rightarrow E = R_1 I_1 + R_2 I_2 \\ \Rightarrow I_2 = \frac{E}{R_1 + R_2}$$

$$\text{car } U = \frac{-R_1 E}{R_1 + R_3} + \frac{R_3 E}{R_1 + R_3}$$

$$-R_1 R_3 + R_1 R_2 + R_3 R_1 + R_1 R_2$$

$$U = \frac{E (R_3 R_2 - R_1 R_4)}{(R_1 + R_3)(R_1 + R_2)}$$

2/ Pour en avoir pour $U=0$ c-à-d $R_3 R_2 = R_1 R_4$

$$\text{donc } R_1 = \frac{R_3 R_2}{R_4}$$

$$\frac{-R_1 R_3 + R_1 R_2}{R_1 + R_3}$$

$$3/ \frac{\Delta U}{\Delta R_1} = \frac{E}{R_1 + R_3} \times \frac{-R_1 R_3 - R_1 R_2}{(R_1 + R_3)^2}$$

$$\text{donc } \Delta R_1 = \frac{(R_1 + R_3)(R_1 + R_2)^2}{E (R_3 R_2 + R_2 R_1)} \Delta U$$

$$\Delta R_1 = \frac{(R_1 + R_3)^2}{E R_2} \Delta U \quad \Delta U = 1 \text{ mV}$$